



PROBLEM SPACE

SOCIAL ROBOT CO-DESIGN CANVASES

What problem are you solving?

USER

Group(s)	Characteristics	Needs
Who are the users? Are there supporting users? For example: students and teachers. Triage Nurses, Emergency Physicians, Charge Nurses primary users	What are the users like? Clinically trained, experienced in patient assessment and emergency medicine, work in fast-paced & high-pressure environments, frequently multitask	What do the users need? Quickly identify high-risk patients, access patient information in one centralized dashboard, receive accurate AI-generated triage recommendations, improve decision-making while maintaining clinical judgement
secondary users Hospital admin, Clinical Support Staff, Patients (indirect users)	Hospital admin monitor departmental performance and resource allocation, Clinical Support Staff assist with patient movement and documentation, Patients seek timely assessment and treatment	Monitor Emergency Department efficiency and patient flow, access reports on waiting times, patient volumen, and resource utilization, streamline communication between healthcare providers and patients

Goal(s)	
What do the primary and secondary users want to accomplish? Complete accurate triage assessments within the recommended timeframe, identify critically ill outpatients as early as possible, receive timely AI-supported recommendations to assist clinical decision-making, prioritize patient safety while maintaining clinician oversight primary users secondary users Monitor patient flow and waiting times in real time, improve communication between healthcare teams, ensure patients receive timely assessment and treatment SHORT-TERM	 Improve consistency and accuracy of triage decisions, reduce the incidence of under-triage and delayed recognition of deteriorating patients, increase confidence in using AI-assisted clinical decision support tools, improve clinical emergency department efficiency. Improve emergency department efficiency, reduce overcrowding and prolonged waiting times, increase patient satisfaction and trust in emergency care and inform future hospital planning and resource allocation. LONG-TERM

ROBOT

Task(s)	
What task does the robot perform? Display the live triage queue; identify high-risk patients; present vital signs and symptoms; generate colour-coded alert states; explain factors contributing to AI risk flags; allow clinician review/override SHORT-TERM	 Integrate EHR/device data; support trend monitoring; improve information sharing across ED; maintain an auditable history of AI recommendations and clinician decisions. LONG-TERM

Advantages			
What advantages does using a robot bring (compared to a computer or human)?			
N/A for screen-based dashboard.		Interface can adapt information to clinical role/workflow.	Consistently calculates/display risk information.
Social skills	User's emotional response	Personalization	Precise tasks
Receives connected vital-sign/device data.	N/A or minimal advantage for desktop HCI.		Integrates with EHR and clinical data sources.
Data collection with sensors	Mobility	Environment manipulation	Connection to systems





ETHICAL CONSIDERATIONS

SOCIAL ROBOT CO-DESIGN CANVASES

Consider potential ethical problems, and potential solutions – both from the user's and robot's perspectives.
Consider the boxes to be guidelines: you don't need to fill each one.

Physical safety

Machines can pinch or crush the user. How is this mitigated?

PROBLEM	USER		SOLUTION
	ROBOT		
Clinicians may miss urgent changes if alerts are unclear during busy triage.		Use highly visible critical alerts and require clinician review of high-risk cases.	
Incorrect or delayed data could create unsafe recommendations.		Flag missing/abnormal data, timestamp information and never autonomously make the final triage decision.	

Data security

Is the robot in a unique data collection position? How is the user's data protected?

PROBLEM	USER		SOLUTION
	ROBOT		
Staff may view sensitive information in a shared ED workspace.		Enforce role-based access, automatic logout and privacy-conscious screen positioning.	
Dashboard processes identifiable clinical and EHR/device data.		Encrypt data, restrict access, maintain audit logs and display only necessary patient information.	

Transparency

How does the robot share an accurate perception of its abilities, intentions and constraints, so the user can evaluate their trust in it?

PROBLEM	USER		SOLUTION
	ROBOT		
Nurses may over-trust an AI risk flag without understanding why it was generated.		Clearly state that CARISURG provides decision support and clinicians retain final authority.	
AI alerts may appear as unexplained risk classifications.		Show contributing factors, relevant vitals and reasons for each risk alert.	

Equality across users

Robots' algorithms can be biased. A robot's appearance could reinforce harmful stereotypes. What are potential issues?

PROBLEM	USER		SOLUTION
	ROBOT		
Alerts may be less reliable for some patient populations.		Encourage clinical review and easy correction/override of recommendations.	
Training data may contain demographic or clinical bias.		Evaluate performance across patient groups, monitor disparities and avoid using protected characteristics as unjustified proxies.	

Emotional consideration

People have been shown to form emotional attachments to robots, as if they were alive. Is this a potential problem?

PROBLEM	USER		SOLUTION
	ROBOT		
Repeated alerts may increase stress or alarm fatigue during busy shifts.		Prioritise alerts by urgency and minimize unnecessary notifications.	
Excessive colours, warnings or sounds could overwhelm clinicians.		Use restrained alert hierarchy with critical notifications reserved for clinically significant risks.	

Behaviour enforcement

People could transfer their inappropriate behaviour, such as rudeness, from robots to humans. How is this mitigated?

PROBLEM	USER		SOLUTION
	ROBOT		
Clinicians may become dependent on AI recommendations or ignore contradictory judgement.		Reinforce that AI supports rather than replaces professional judgement.	
Interface design could unintentionally push users toward accepting its recommendation.			





DESIGN GUIDELINES

SOCIAL ROBOT CO-DESIGN CANVASES

What things are important to consider in the robot's design?

Advantage guidelines

What advantages can the robotic solution have?
Think back to what you defined in the solution space canvas.

- Reduce repetitive review of patient data.
- Highlight high-risk patients rapidly.
- Consolidate symptoms, vitals and risk information in one view.
- Support faster, more focused clinical assessment.
- Preserve clinician control over final triage decisions.

Ethical guidelines

What ethical considerations does the robot have?
Think back to what you defined in the ethics canvas.

- AI recommendations must remain advisory.
- Show reasons behind risk alerts.
- Allow nurses to modify or override recommendations.
- Protect identifiable patient information.
- Monitor for bias across patient groups.
- Clearly distinguish AI-generated information from clinician decisions.

ROBOT DIMENSIONS

Environment guidelines

What should the robot's context be like?

For example:

- If users are especially vulnerable, should it optimize for support?
- If the robot is part of a strict process, should it optimize for efficiency and security?

- AI recommendations must remain advisory.
- Show reasons behind risk alerts.
- Allow nurses to modify or override recommendations.
- Protect identifiable patient information.
- Monitor for bias across patient groups.
- Clearly distinguish AI-generated information from clinician decisions.

Form guidelines

What guides the design of the robot's outward qualities?

For example:

- Should the robot be designed to appear especially approachable, or more industrial?
- Should it be simple, or detailed?

- Use a clean, professional clinical-dashboard appearance.
- Maintain consistent layout and visual hierarchy.
- Use large, legible text and controls.
- Clearly differentiate alert states without relying on colour alone.
- Avoid decorative elements that distract from clinical information.

Interaction guidelines

What guides the design of interaction?

For example:

- Is the interaction multimodal, or is one modality optimized for efficiency?
- Should the user feel empowered and lead the interaction, or does the robot provide safety via leadership?
- Is the goal of the interaction to complete a task, or explore?

- Prioritise queue → patient → alert → action workflow.
- Allow rapid access to symptoms, vitals and AI reasoning.
- Provide clear review, modify and override controls.
- Minimise data re-entry.
- Require acknowledgement of critical alerts where appropriate.
- Keep essential functions within minimal navigation steps.

Behaviour guidelines

What guides the design of the robot's behaviour?

For example:

- Should behaviour be simple, or sensitive to context?
- Does the robot have internal drivers, or does it react to external stimuli?
- Does the robot have social skills?

- Update patient status when new information is received.
- Escalate clinically significant alerts without excessive notifications.
- Never independently assign the final triage decision.
- Clearly indicate missing, uncertain or outdated data.
- Record clinician review/override actions.
- Avoid repeatedly prompting clinicians after a decision is made.





ROBOT DESIGN MVP

SOCIAL ROBOT CO-DESIGN CANVASES

It's time to design your robot MVP (Minimum Viable Product)! Remember the guidelines you defined.

Where and when

Emergency Department triage/workstation area, Used continuously during patient intake and triage, particularly when nurses are reviewing newly collected patient information and prioritising the waiting queue.

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

See mockup design

Robot's role

Is the robot a friend? Teacher? Helper?
Clinical decision-support assistant that organises pre-triage information, highlights potential high-risk patients and supports nurse prioritisation without replacing clinical judgement.

Personality

Does the robot have specific characteristics?
Does it have emotional states, or needs?
Professional, calm, concise and non-alarming. Communicates urgency clearly without appearing authoritative or replacing clinician expertise.

Context-based behaviour

What external and environmental factors affect behaviour?
What data is used to adapt to context?
Prioritises information according to patient risk, updates when new symptoms/vitals are received, escalates significant alerts and identifies missing or uncertain data.

Connection to systems

Is the robot connected to external systems, such as software, data, other robots?
Receives data from the CARISURG pre-triage robot, connected vital-sign devices and potential EHR sources. Sends organised patient information and alerts to the nurse-facing dashboard.

Interaction modalities

What modalities are inputs to the robot? What modalities does the robot output?

INPUT

☐ voice ☒ touch ☒ screens ☐ lights ☒ other keyboard/mouse + clinical data feeds
☐ sounds ☐ smell ☒ other ☐ gestures ☐ facial expressions

OUTPUT

☐ voice ☐ touch ☒ screens ☐ lights ☒ other Visual alerts/notifications
☐ sounds ☐ smell ☐ other ☐ gestures ☐ facial expressions

Interaction flow

Describe the most important interaction of the robot.

Note: only fill the bottom row if your robot is teleoperated.

	BEFORE	DURING	AFTER
USER	Opens dashboard and views triage queue.	Reviews patient symptoms, vitals, AI risk flag and explanation; confirms/clarifies patient information.	Makes final triage decision and proceeds with appropriate clinical action.
ROBOT	Receives and organizes incoming pre-triage data.	Displays patient risk status, relevant data and explainable alerts; allows review/override.	Records clinician action, updates patient status and maintains audit information.
ROBOT OPERATOR (optional)			





ENVIRONMENT

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What is the robot's context of operation?
You can use the "Ecosystem" canvas to dive deeper into this topic.

Where

What place?
Does it change?

Emergency Department triage area/nursing workstation. Used on a clinical computer within a busy, shared workspace with frequent interruptions, variable lighting and limited screen space.

User(s)

Who is using the robot?

Primary: Triage nurses using CARISURG to review pre-collected symptoms, vital signs, risk alerts and the patient queue.

When

What time of day?
Does it change?

Used continuously during ED operating hours, especially during patient arrival and formal triage. Workload and patient volume may vary significantly throughout a shift.

Secondary user(s)

Are there secondary users?
E.g. teachers that help students use a robot.

Emergency physicians, charge nurses and authorised ED staff who may review patient status, triage information or escalation alerts.

Data collection

Does the robot collect data from its environment?
How is it stored?

Receives demographics, chief complaint, symptoms, vital signs, arrival information and pre-triage data. Data may originate from the HRI robot, connected devices, manual entry and EHR sources. Access and storage must follow hospital privacy/security controls.

TRADE-OFF:

More data collection requires more attention to data security.

Simultaneous users

How many users should be able to use the robot simultaneously?



TRADE-OFF:

More simultaneous users requires a more sophisticated robot.



Multiple authorised clinicians may need to access patient information simultaneously from different workstations.

External sensors and actuators

Does the robot use external sensors?
Does it have external actuators, such as lights or limbs?

No direct sensors required at the nurse workstation. System receives measurements from connected vital-sign devices and HRI robot sensors through integrated data feeds.

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

Connects to the CARISURG AI system and the hospital EHR.

Supports secure exchange of patient information, alerts and clinician actions.





FORM

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What are the robot's outward qualities? If an existing robot is used, are its qualities modified?

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

See mockup design

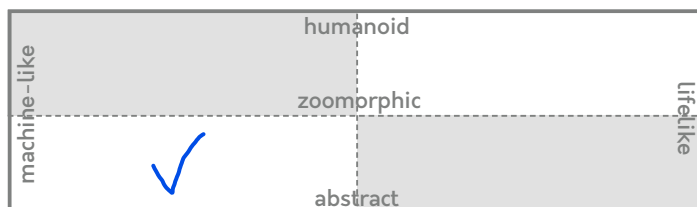
Appearance

Is the robot more machine or lifelike? Is it human-shaped, animal-shaped, or abstract?



TRADE-OFF:

Robots that appear more human and lifelike are expected to be more sophisticated in features.



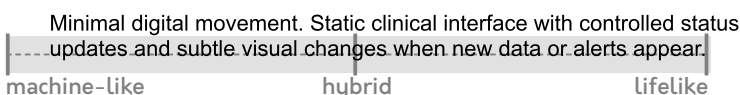
Size

How big is the robot?
Desktop/laptop clinical display; responsive to standard hospital workstation screens.



Character of movement

What is the robot's movement like?



Voice & sounds

Does the voice have a gender or an age? What are pitch, speed and prosody like? Is the voice always the same?
Does the robot make sounds: music, "beep"s, animal noises?
When are these sounds heard?

Primarily silent. There are optional brief audio cues for critical alerts only; routine notifications remain visual to minimize alarm fatigue.

Mobility

Does the robot move across space? Does it move in place?

No physical mobility. Web-based interface may be accessed from authorized clinical workstations within the ED.

Visual cues

Does the robot have expressions, lights, a screen or other visual elements?

Clear patient queue, readable typography, risk-status indicators, vital-sign displays and alert icons. Critical states use colour plus text/icons, so meaning does not depend on colour alone.

Touch & smell sensations

Is the robot soft or rough, warm or cold?
How does the robot smell?
Touch and smell are especially important in close interactions.

Not applicable to smell. Interaction occurs through standard workstation controls or touchscreen where available; controls should be large enough for rapid, accurate selection.

